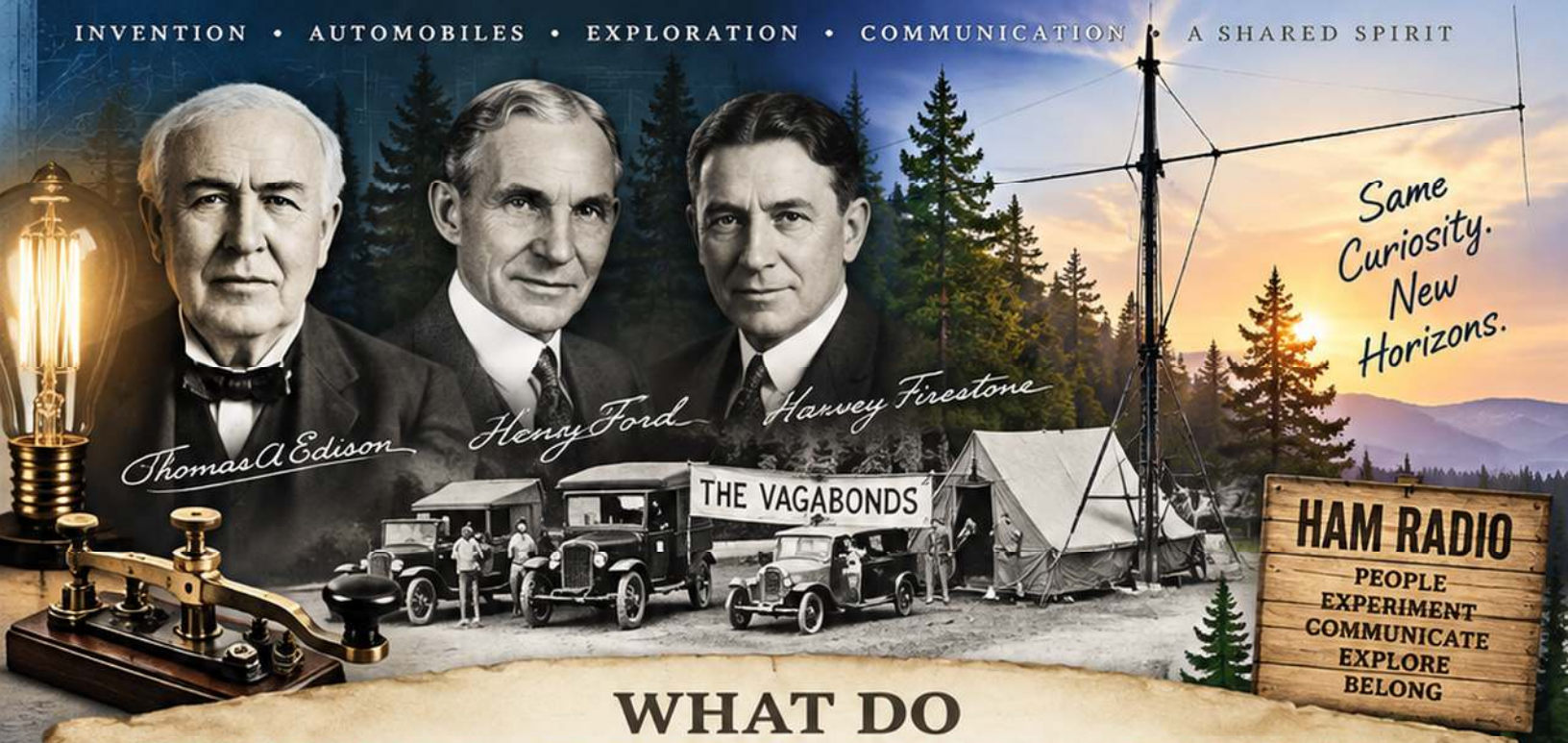


WHAT DO EDISON, FORD, FIRESTONE, AND HAM RADIO HAVE IN COMMON?

From Dots and Dashes to a Brighter World

DIFFERENT TIMES. SAME SPIRIT.

RADIO CONNECTS GENERATIONS

THOMAS EDISON.

Henry Ford.

Harvey Firestone.

And amateur radio.

At first glance, one of these things doesn't seem to belong.

Edison helped electrify the world. Ford put America on wheels. Firestone helped put tires under those wheels.

And amateur radio?

Well, we put antennas in trees.

So what could these possibly have to do with one another?

As it turns out, quite a bit.

And the trail begins with something every CW operator will recognize.

Before Edison Was the Wizard

Long before Thomas Edison became famous for electric lights, phonographs and motion pictures, he was a telegraph operator.

That little detail turns out to be important.

Edison learned telegraphy as a teenager and eventually worked professionally as a telegraph operator. Electricity and communications became central themes of his early career.

And Morse code wasn't something Edison simply left at work.

His first child, Marion Estelle Edison, was nicknamed "Dot," a direct reference to the dots and dashes used in telegraphy. Her younger brother Thomas Alva Edison Jr. became "Dash."

Think about that for a moment.

Plenty of amateur radio operators love CW.

We put Morse on coffee mugs.

We put it on T-shirts.

We put our callsigns in Morse on shack signs.

Thomas Edison named the kids after it.

That's commitment.

But the best Edison Morse story was still to come.

The World's Shortest-Range Communications System

After Edison's first wife, Mary, died, Edison met a young woman named Mina Miller.

Soon Edison encountered a communications problem that generations of young couples would later solve with text messages:

How do you have a private conversation when the family is sitting right there?

Edison had a solution.

He taught Mina Morse code.

According to the National Park Service, Edison said that he and Mina could then communicate secretly even while members of the family were watching.

And apparently the system worked rather well.

Because Edison claimed that one day he tapped a question into Mina's hand.

Would she marry him?

Mina tapped back:

Yes.

There was even a witness.

Edison's daughter Marion, appropriately enough the child nicknamed Dot, later claimed that she saw her father propose to Mina using Morse code.

So, depending upon how generous we're feeling with our terminology, Edison had developed a private communications system.

No repeater.

No cell service.

No internet.

No encryption.

And absolutely no monthly subscription.

Admittedly, the effective range was only a few inches.

But the message got through.

Morse: 1.
Nosy relatives: 0.

And if you're wondering whether that counts as a QSO, we'll leave that argument for the next club meeting.

Morse Was More Than a Parlor Trick

This is where our story starts becoming more interesting.

Telegraphy wasn't merely Edison's first job. It influenced the way he thought about communications and invention.

When Edison was developing the phonograph in 1877, his notes actually compared recording the human voice with recording telegraphic signals.

The National Park Service documents Edison thinking about telephone signals as a kind of "speaking telegraph." Telegraph dots and dashes could already be recorded and replayed mechanically. Edison began considering whether human speech might be captured and reproduced in a similar fashion.

So the telegraph wasn't simply a colorful chapter from Edison's youth.

It helped shape the thinking that led to some of his later inventions.

But we still haven't explained Henry Ford and Harvey Firestone.

For that, we're going camping.

Enter Ford and Firestone

Henry Ford became a close friend of Thomas Edison.

Add Harvey Firestone, founder of the Firestone Tire and Rubber Company, and naturalist and writer John Burroughs, and we suddenly have a rather remarkable collection of friends.

An inventor.

An automobile manufacturer.

A tire manufacturer.

A naturalist.

What do successful men like these do when they want to escape civilization?

Apparently they load an unreasonable quantity of equipment into vehicles and head into the woods.

Any amateur radio operator who has ever participated in Field Day should immediately recognize the symptoms.

Meet the Vagabonds

From 1915 into the 1920s, Edison, Ford, Firestone and Burroughs embarked on a series of automobile camping adventures.

They became known as the Vagabonds.

And calling what they did "camping" requires a somewhat flexible interpretation of the word.

As the expeditions grew, so did the operation.

There were automobiles.

Trucks.

Tents.

Equipment.

Food.

Support personnel.

People to maintain the vehicles.

People to prepare meals.

Photographers and reporters.

The Henry Ford's historical collection even contains photographs of their 1921 caravan of trucks and automobiles.

Let's review.

A group of technically minded people...

loads several vehicles with far more equipment than any normal person believes necessary...

drives into the countryside...

sets up a temporary camp...

plays with technology...

eats...

talks...

occasionally naps...

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and attracts curious people wondering what they're doing.

Ladies and gentlemen, we are getting dangerously close to Field Day.

They were basically one antenna launcher away from needing an ARRL section abbreviation.

They Even Came Through Pennsylvania

Now this gets a little closer to home.

During the 1918 Vagabonds expedition, Ford met Firestone in Ohio and then joined Edison and Burroughs in Pennsylvania. From there the group traveled through West Virginia, Virginia, Tennessee and North Carolina.

So picture the scene.

A caravan of vehicles loaded with equipment.

Thomas Edison is there.

Henry Ford is there.

Harvey Firestone is there.

They're traveling through Pennsylvania.

If someone had handed Edison a transmitter and said, "Tom, see if you can work New Jersey," amateur radio history might have become considerably more interesting.

Unfortunately, we don't have evidence that the Vagabonds were operating an amateur station on these expeditions.

So we can't call it Field Day.

But we can certainly accuse them of suspiciously Field-Day-like behavior.

But Where Is the Radio?

Up to this point we have a wonderful coincidence.

Edison was a telegrapher.

He knew Morse.

He used Morse at home.

He used ideas originating in telegraphy while thinking about other communications technologies.

And he went on elaborate camping trips with Ford and Firestone.

That's enough for a fun story.

But it isn't the real punch line.

Because Thomas Edison has a much more direct connection to radio technology.

And strangely enough, it starts with a light bulb.

The Edison Effect

In 1883, while Edison's laboratory was experimenting with incandescent lamps, something peculiar was observed.

A current could flow through the vacuum inside a lamp between the heated filament and another electrode.

The phenomenon became known as the Edison Effect.

Edison made some practical use of it, but its greater significance wasn't yet understood.

Then along came British physicist and electrical engineer John Ambrose Fleming.

Fleming investigated the phenomenon and eventually realized that it could be used to detect radio signals.

In 1904 he developed the thermionic valve.

The Smithsonian's National Museum of American History describes Fleming's device as the first radio tube and directly traces its development to the Edison Effect.

Now our little camping story has wandered straight into the radio shack.

Follow the path:

Edison the telegrapher

☐

Morse code and electrical communication

☐

Edison's electrical experimentation

☐

The Edison Effect

☐

Fleming's thermionic valve

☐

Practical radio tubes and electronic circuits

☐

Radio

☐

Amateur Radio

Suddenly the connection isn't just a joke.

There's an actual technological family tree.

Ford and Firestone Weren't Finished Either

Years later, our three friends would come together for something much more serious than camping.

In 1927, Edison, Ford and Firestone were concerned about America's dependence on foreign sources of natural rubber.

That was a significant problem.

Ford needed rubber for automobiles.

Firestone obviously needed enormous quantities of it for tires.

And Edison?

Edison did what Edison did.

He started experimenting.

The three men formed the Edison Botanic Research Corporation.

The following year a research laboratory was built in Fort Myers, Florida.

The goal was straightforward but enormously ambitious:

Find a plant that could be grown quickly in the United States and provide a practical domestic source of rubber.

Under Edison's leadership, researchers tested more than 17,000 plant samples.

Eventually Edison selected goldenrod as the most promising candidate.

Think about that.

Seventeen thousand samples.

Different materials.

Different experiments.

Different results.

Try something.

Measure it.

Change something.

Try again.

Keep what works.

Discard what doesn't.

Repeat.

Now that should sound familiar to amateur radio operators.

Except when a ham tries 17,000 different antenna configurations, we call it:

Saturday.

Maybe That's the Real Connection

Thomas Edison, Henry Ford and Harvey Firestone weren't the founders of amateur radio.

We're not going to rewrite history and issue them honorary callsigns.

But perhaps that's not really the point.

They lived during an extraordinary period when electricity, automobiles, telecommunications and radio were

transforming everyday life.

They were curious.

They experimented.

They built things.

They took things apart.

They identified problems and tried to solve them.

Sometimes their ideas worked.

Sometimes they didn't.

And occasionally they packed an absurd amount of equipment into automobiles and disappeared into the countryside with their friends.

More than a century later, amateur radio operators are still doing many of those same things.

We experiment with antennas.

We build equipment.

We troubleshoot problems that nobody else knew existed.

We communicate using technology whose ancestry stretches all the way back to the telegraph.

We still use Morse code.

And every June, thousands of us pack radios, coax, antennas, batteries, generators, laptops, food, tents, tables, chairs and approximately fourteen adapters that nobody remembers packing into vehicles and head into the field.

Maybe Edison, Ford and Firestone weren't hams.

But I suspect they'd understand us.

Edison would recognize the Morse.

Ford would approve of the mobile installation.

Firestone would make sure we got there.

And if the Vagabonds walked into a modern Field Day site and saw the trucks, tents, generators, food, wires, equipment and people enthusiastically experimenting with technology, they might feel strangely at home.

They'd probably have only one question:

"What's the antenna for?"

And after explaining that we're trying to talk to someone halfway around the world using a radio powered by a generator in the middle of a field, Edison would probably want to see the schematic.

Ford would want to know how everything fit in the truck.

Firestone would check the tires.

And eventually someone would ask the most important question heard at Field Day:

"Has anybody eaten yet?"

Because apparently some technologies really haven't changed very much.

73

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